🔑 **Answer Key — Teacher Copy · Lesson 2-12**

MSTAR PRACTICE WORKSHEET

Lesson 2-12: Multiply Decimals

Grade 6 Mathematics · Standard 6.NOS.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is 3.4 × 2.6?

A. 0.884

B. 8.48

C. 8.84

D. 88.4

34 × 26 = 884. There are 2 total decimal places (1 + 1), so place the decimal to get 8.84.

Question 2 SELECTED RESPONSE — CORRECT: B

What is 0.7 × 1.5?

A. 0.105

B. 1.05

C. 1.50

D. 10.5

7 × 15 = 105. There are 2 total decimal places (1 + 1), so place the decimal to get 1.05.

Question 3 SELECTED RESPONSE — CORRECT: C

What is 5.3 × 4?

A 2.12

B 20.12

C 21.2

D 212

53 × 4 = 212. There is 1 total decimal place, so place the decimal to get 21.2.

Question 4 SELECTED RESPONSE — CORRECT: B

Which estimate is closest to 4.8 × 3.2?

A 4 × 3 = 12

B 5 × 3 = 15

C 5 × 4 = 20

D 48 × 32 = 1,536

Round 4.8 to 5 and 3.2 to 3. The estimate 5 × 3 = 15 is closest to the exact answer 15.36.

Question 5 SELECTED RESPONSE — CORRECT: C

A car travels 55.5 miles per hour for 2.4 hours. How far does it go?

A 13.32 miles

B 57.9 miles

C 133.2 miles

D 1,332 miles

555 × 24 = 13,320. Total decimal places: 1 + 1 = 2. Answer: 133.20 = 133.2 miles.

Question 6 SELECTED RESPONSE — CORRECT: C

A customer orders 3 taco baskets at $4.25 each. What is the total before tax?

A $1.275

B $12.15

C $12.75

D $127.50

3 × 425 = 1,275. The factor $4.25 has 2 decimal places, so 3 × $4.25 = $12.75.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Find 2.4 × 0.5.

A 0.12

B 1.2

C 12

D 120

Multiply as whole numbers: 24 × 5 = 120. The factors have 1 + 1 = 2 decimal places in total, so the product is 1.20, which is 1.2.

- **A:** The point moved three places instead of two. Count one place in 2.4 and one in 0.5.
- **C:** 24 x 5 = 120 is the whole-number product. The two factors hold 2 decimal places between them, so the point moves two places left.
- **D:** That is the whole-number product 24 x 5 with no decimal point placed at all.

Part B — correct answer: C

How is the number of decimal places in the product determined?

A Use the number of decimal places in the larger factor.

B Multiply the number of decimal places in the two factors.

C Add the number of decimal places in the two factors.

D The product always has one decimal place.

2.4 has 1 decimal place and 0.5 has 1, so the product has 1 + 1 = 2 decimal places before trailing zeros are dropped.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about multiplying decimals:

A Multiply the factors as if they were whole numbers first.

B The total number of decimal places in the factors gives the number in the product.

C Multiplying by a number less than 1 gives a product smaller than the other factor.

D The decimal points must be lined up before multiplying.

E Estimating first is a good way to check the placement of the decimal point.

A, B, C, and E are true. D is false — lining up decimal points is required for ADDING and subtracting, not for multiplying.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A driver buys 9.5 gallons of gas at $3.40 per gallon. Show how to find the total cost. Use estimation to check if your answer is reasonable.

Model answer: Total cost is gallons times price per gallon, so the setup is 9.5 × $3.40. Multiplying 95 × 34 = 3,230 and giving the product 1 + 1 = 2 decimal places gives $32.30. Rounding to 10 gallons at about $3 each predicts roughly $30, so $32.30 is reasonable.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
